

Applied MSc in Data Science & Artificial Intelligence
Applied MSc in Data Engineering for Artificial Intelligence
Executive MSc in Artificial Intelligence
Digital Industry & AI

“Artificial Neural Networks with Benoit Mialet”

Volume of classes hours: 25 hrs ($\pm$ same personal work expected)

Course summary:

❖ Theoretical part:

- Introduction to neural networks
- The ANN: Perceptron, layers, weights, biases, hyperparameters
- Activation and cost functions
- Optimization algorithms
- Backpropagation
- The learning mechanism
- Classification and regression

❖ Practical part:

- Introduction to PyTorch 2.0
- Neural network training and evaluation using PyTorch and Tensorboard
- Hyperparameter tuning using Optuna
- Real-life case-study regression and classification examples

Course Objectives:

- Acquire a thorough understanding of the concepts at the base of neural networks
- Acquire hands-on practice of PyTorch and Google Colab

Theoretical background used:

- Differential calculus
- Python
- Machine Learning
- Continuous Optimization

Technologies Used:

Python, PyTorch 2.0, Tensorboard, Optuna, Scikit-learn, Numpy, Seaborn, Matplotlib, Google Colab

Course evaluation:

Students will be evaluated through a Multiple-Choice Question (MCQ) exam that reprises foundational concepts seen in class.